Evaluating Methods for Choosing K in K-Means Clustering: A comparative Study of Elbow and Silhouette Techniques

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Question:
How can we determine the optimal number of clusters in K-Means, and how reliable are the Elbow method and Silhouette score across datasets with different structures?

1. Introduction:

Clustering is an unsupervised machine learning technique used to group data points based on similarity. Among clustering algorithms, **K-Means** is widely used due to its simplicity and efficiency. It partitions data into **K clusters**, where each point belongs to the cluster with the nearest centroid. A critical challenge is choosing the correct number of clusters (**K**), as a poor choice can lead to misleading insights.

This report investigates two methods to determine K: the **Elbow method** and the **Silhouette score**. The Elbow method evaluates the Within-Cluster Sum of Squares (WCSS) to find the point beyond which adding more clusters yields diminishing returns. The Silhouette score measures how similar points are within a cluster compared to other clusters, providing insight into cluster cohesion and separation.

We analyse three datasets: **well-separated clusters** (ideal scenario), **overlapping clusters** (challenging scenario), and **non-spherical clusters** (violates K-Means assumptions). The aim is to compare these methods and assess their reliability across different cluster structures.

2. K-Means Overview

K-Means is an iterative algorithm that assigns points to the nearest cluster centroid and updates centroids until convergence. The objective is to minimize **Within-Cluster Sum of Squares (WCSS):**

$$WCSS = \sum_{i=1}^{K} \sum_{x \in C_i} \| x - \mu_i \|^2$$

where $\mu_i$is the centroid of cluster C_i. Distance is measured using Euclidean distance:

$$d(x, \mu) = \sqrt{\sum_j (x_j - \mu_j)^2}$$

3. Methods for Choosing K:

Elbow Method:

- Plot WCSS against K
- Identify the “elbow” where adding more clusters yields diminishing returns
- Quick visual interpretation

Silhouette Score:

- Measures cohesion and separation
- Average score closer to 1 → well-defined clusters
- Provides quantitative guidance for overlapping clusters

Both methods provide complementary insights: Elbow is visual and intuitive, Silhouette is numeric and precise.

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4. Experimental Setup:

Datasets:

- **Well-separated clusters:** ideal for K-Means
- **Overlapping clusters:** more challenging
- **Non-spherical clusters:** violates K-Means assumptions

Implementation:

- Generated datasets using make_blobs and make_moons
- Applied K-Means with K ranging from 2 to 6
- Calculated WCSS and Silhouette scores
- Visualized clusters and evaluation plots using matplotlib

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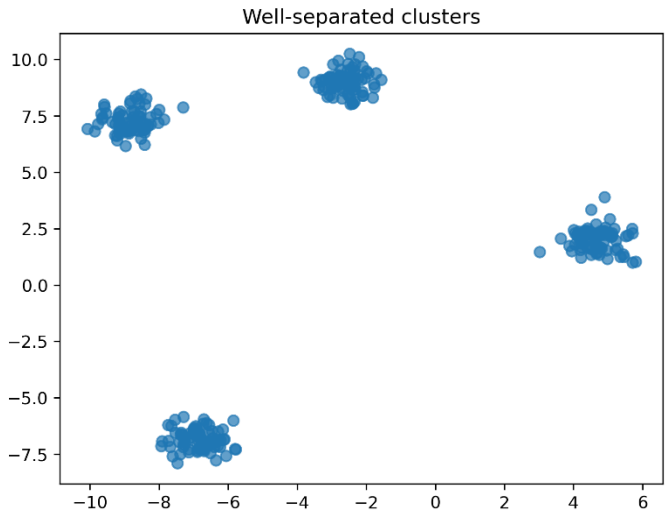
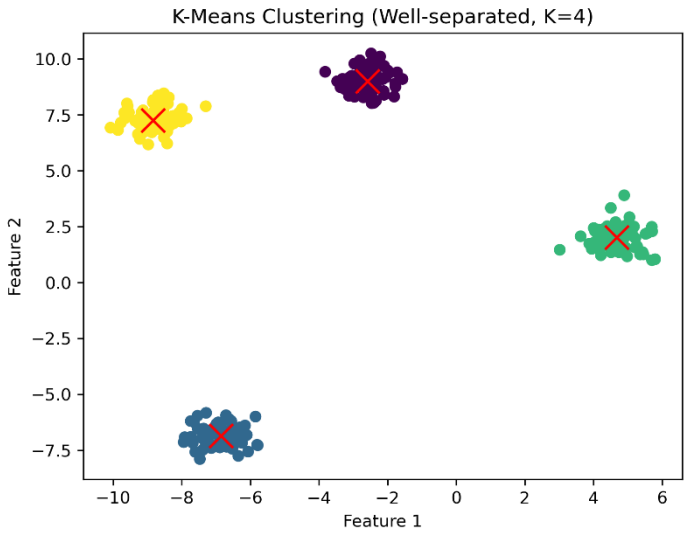
5. Results:

a) Well-Separated Clusters:

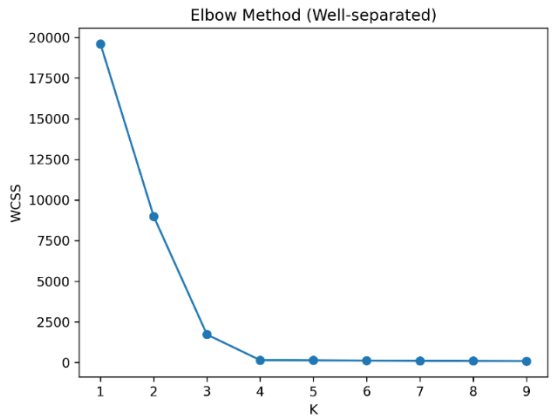
- Scatter plot:



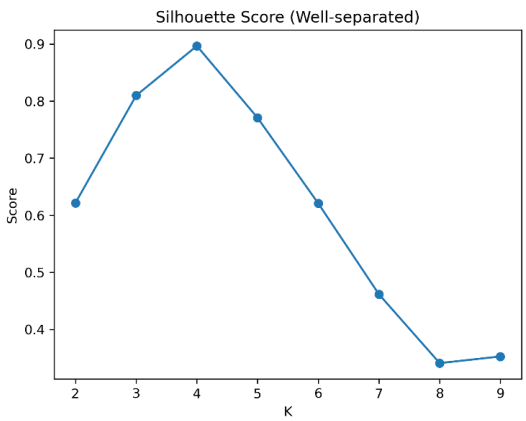
K-Means clustering:



- Elbow plot:



Silhouette plot:

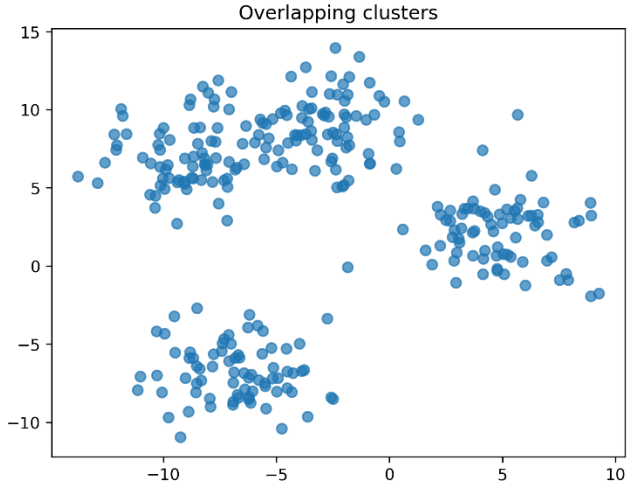


Observations:

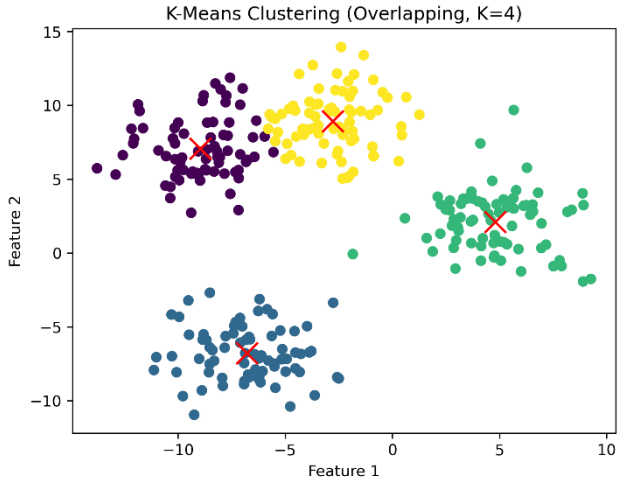
- Clusters are clearly separated
- Both Elbow and Silhouette methods correctly suggest K=4

b) Overlapping Clusters

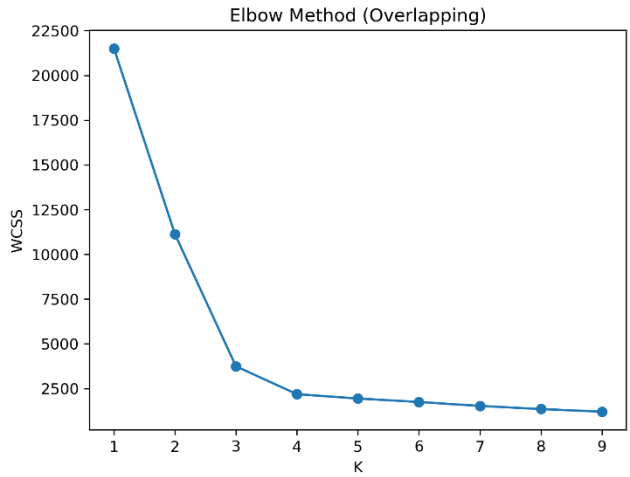
- Scatter plot:



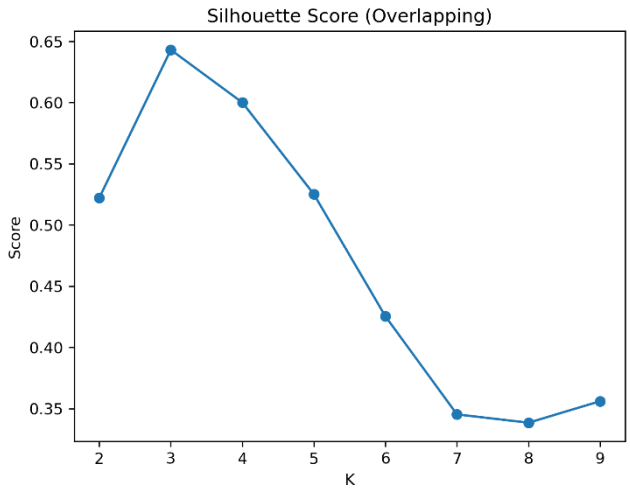
K-Means clustering:



- Elbow plot:



Silhouette plot:

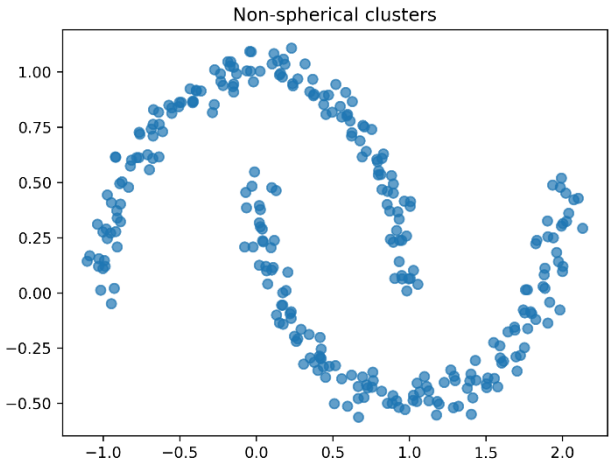


Observations:

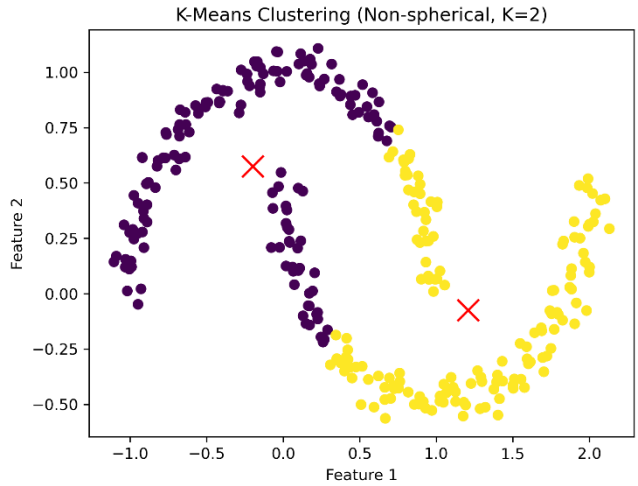
- Boundary points are misclassified
- Elbow method less clear
- Silhouette score more reliable

c) Non-Spherical Clusters

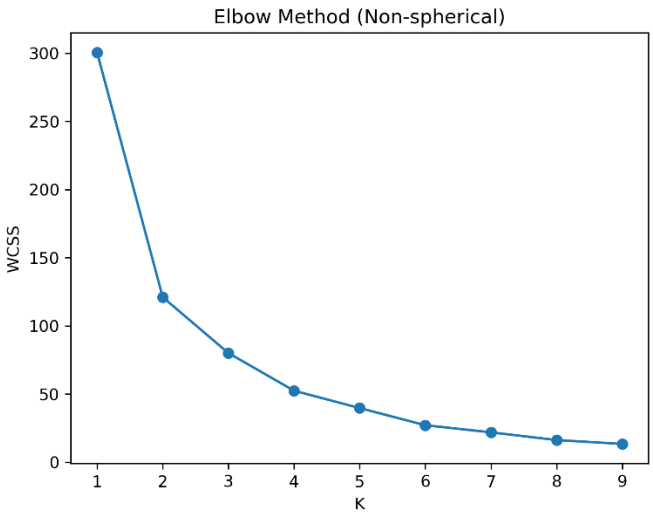
- Scatter plot:



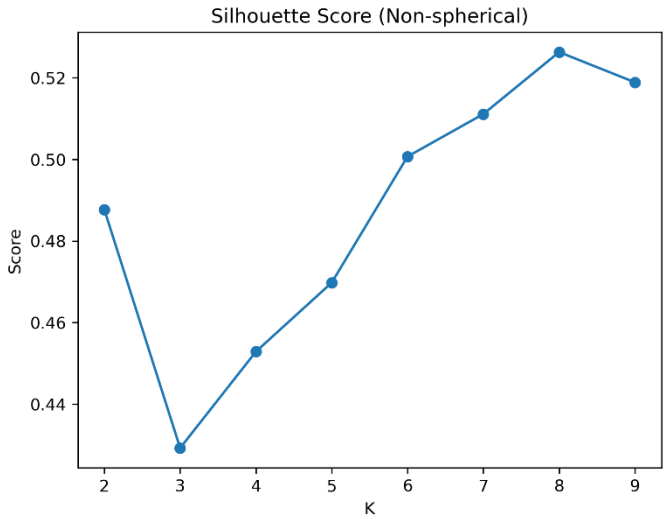
K-Means clustering:



- Elbow plot:



Silhouette plot:



Observations:

- K-Means fails to capture curved shapes
- Both methods unreliable for selecting K

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6. Critical Analysis / Discussion

- **Elbow Method:** geometric, simple, effective for well-separated clusters
- **Silhouette Score:** quantitative, better for overlapping clusters
- Dataset structure affects method reliability
- **Limitations of K-Means:** assumes spherical clusters, sensitive to outliers, struggles with varied cluster sizes or densities

Recommendations:

- Use Silhouette score when clusters overlap
- Inspect data visually before applying K-Means

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7. Conclusion

- Well-separated clusters → both methods accurate
- Overlapping clusters → Silhouette preferred
- Non-spherical clusters → K-Means unsuitable

Takeaway: Always inspect data and use multiple evaluation methods when choosing K. Alternative clustering methods may be necessary for non-spherical or complex data.

Here is my Github link where you can find all the plots, code - Mygithub

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